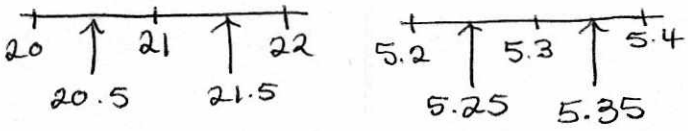
1 A rectangle has a length of 21cm, to the nearest cm, and a width of 5.3cm, to the nearest mm.

(a) Work out the upper bound for the perimeter of the rectangle.



$2(21.5) + 2(5.35)$

53.7 cm

(2)

(b) Work out the lower bound for the area of the rectangle.

20.5×5.25

107.625 cm^2

(2)

(Total for question 1 is 4 marks)

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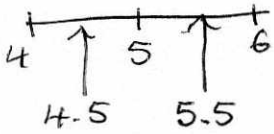
2 A circle has a radius of 5cm, to the nearest cm.

(a) Work out the lower bound for the circumference of the circle.
Give your answer in terms of π.



$2\pi(4.5)$

9π cm

(2)

(b) Work out the upper bound for the area of the circle.
Give your answer in terms of π.

$\pi(5.5)^2$

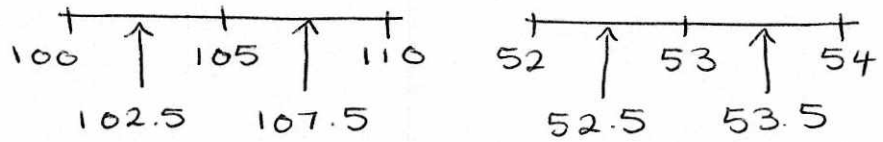
30.25π cm^2

(2)

(Total for question 2 is 4 marks)

3 A rectangular field has a length of 105 metres, to the nearest 5 metres, and a width of 53 metres, to the nearest metre.

(a) Work out the lower bound for the perimeter of the field.



$2(102.5) + 2(52.5)$

310 m

(2)

(b) Work out the upper bound for the area of the field.

107.5×53.5

5751.25 m^2

(2)

(Total for question 3 is 4 marks)

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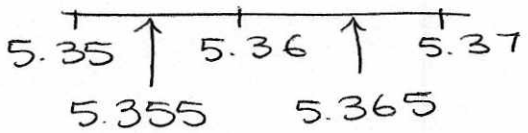
4 A circle has a radius of 5.36cm, correct to 2 decimal places.

(a) Work out the lower bound for the circumference of the circle.

Give your answer to 2 decimal places.



$2\pi(5.355)$

33.65 cm

(2)

(b) Work out the upper bound for the area of the circle.

Give your answer to 3 significant figures.

$\pi(5.365)^2$

90.4 cm^2

(2)

(Total for question 4 is 4 marks)

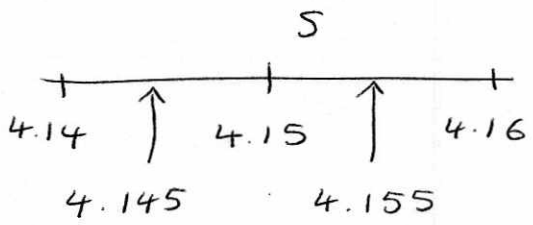
5

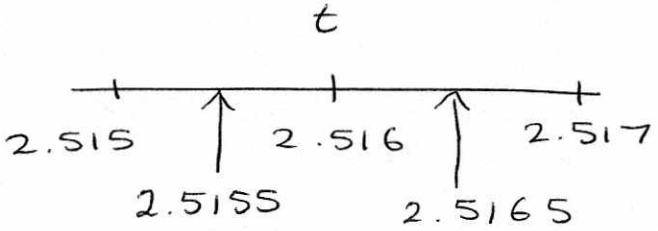
$$v = \frac{s}{t}$$

s = 4.15 correct to 2 decimal places
t = 2.516 correct to 3 decimal places

Work out the upper bound for v.
Give your answer to 3 decimal places.

$$\text{upper } v = \frac{\text{upper } s}{\text{Lower } t}$$





$$\frac{4.155}{2.5155}$$

1.652

(Total for question 5 is 3 marks)

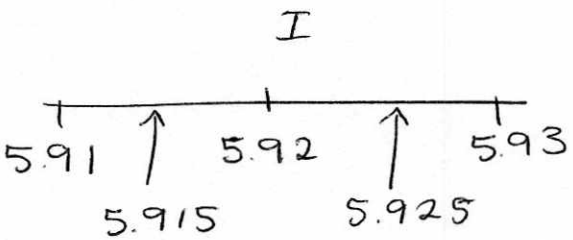
6

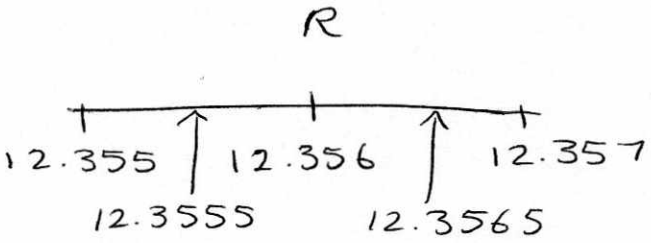
$$V = IR$$

I = 5.92 correct to 2 decimal places
R = 12.356 correct to 3 decimal places

Work out the upper bound for V.
Give your answer to 3 decimal places.

upper V = upper I × upper R



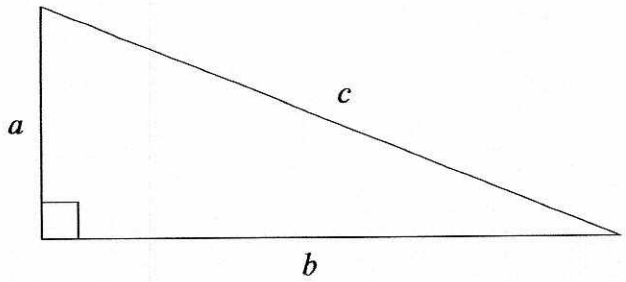


5.925 × 12.3565

73.212

(Total for question 6 is 3 marks)

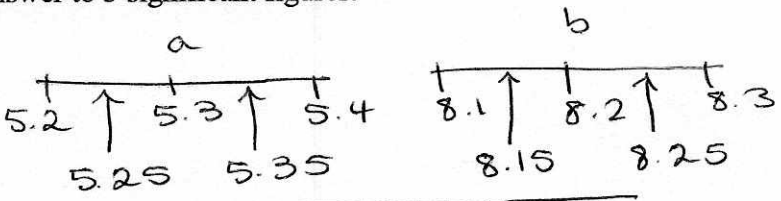
7



a = 5.3 cm correct to the nearest mm
b = 8.2 cm correct to the nearest mm

$a^2 + b^2 = c^2$

Calculate the lower bound for *c*.
You must show all your working.
Give your answer to 3 significant figures.

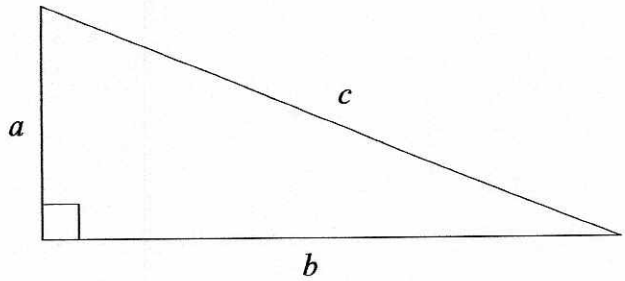


$\sqrt{(5.25)^2 + (8.15)^2}$

9.69 .....cm

(Total for question 7 is 4 marks)

8

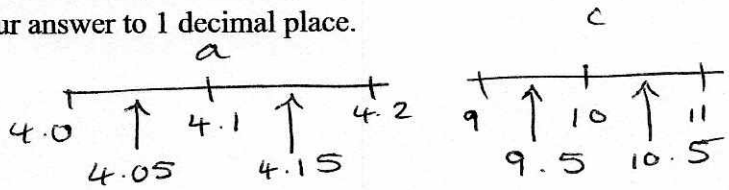


a = 4.1 cm correct to the nearest mm
c = 10 cm correct to the nearest cm

$b^2 = c^2 - a^2$

$(\text{Lower } b)^2 = (\text{Lower } c)^2 - (\text{upper } a)^2$

Calculate the lower bound for *b*.
You must show all your working.
Give your answer to 1 decimal place.



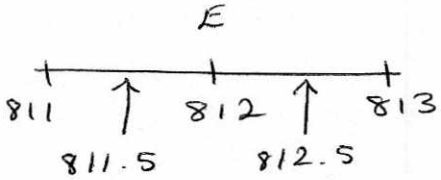
$\sqrt{(9.5)^2 - (4.15)^2}$

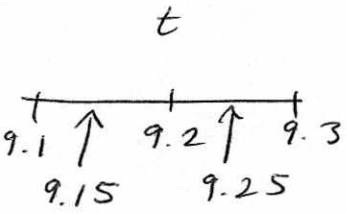
8.5 .....cm

(Total for question 8 is 4 marks)

9 $P = \dfrac{E}{t}$ $E = 812$ correct to 3 significant figures
$T = 9.2$ correct to 1 decimal place

By considering bounds, work out the value of P to a suitable degree of accuracy.
Give a reason for your answer.





upper $P = \dfrac{\text{upper } E}{\text{lower } t}$

$= \dfrac{812.5}{9.15}$

$= 88.79781421$

lower $P = \dfrac{\text{lower } E}{\text{upper } t}$

$= \dfrac{811.5}{9.25}$

$= 87.72972973$

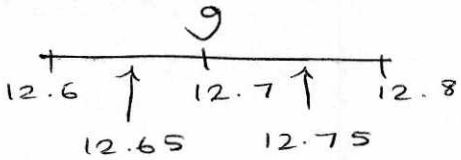
Both round to 90 (1sf)

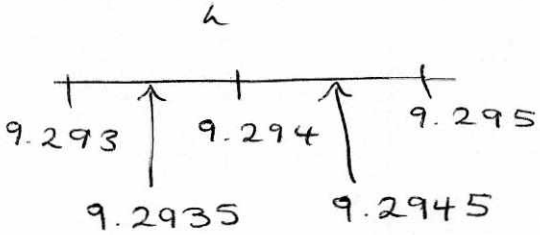
90

(Total for question 9 is 5 marks)

10 $f = \dfrac{\sqrt{g}}{h}$ $g = 12.7$ correct to 3 significant figures
$h = 9.294$ correct to 3 decimal places

By considering bounds, work out the value of f to a suitable degree of accuracy.
Give a reason for your answer.





upper $f = \dfrac{\sqrt{\text{upper } g}}{\text{lower } h}$

$= \dfrac{\sqrt{12.75}}{9.2935}$

$= 0.3842163037$

~~upper~~ lower $f = \dfrac{\sqrt{\text{lower } g}}{\text{upper } h}$

$= \dfrac{\sqrt{12.65}}{9.2945}$

$= 0.38266543 1$

Both round to 0.38 (2dp / 2sf)

0.38

(Total for question 10 is 5 marks)

11

F = 25.14 N correct to 2 decimal places
A = 4.29 m^2 correct to 3 significant figures

$$p = \frac{F}{A}$$

p = pressure
F = force
A = area

By considering bounds, work out the value of p to a suitable degree of accuracy.
Give a reason for your answer.

F

25.13 ↑ 25.14 ↑ 25.15

25.135 25.145

A

4.28 ↑ 4.29 ↑ 4.30

4.285 4.295

upper p = $\frac{\text{upper } F}{\text{lower } A}$

$= \frac{25.145}{4.285}$

$= 5.868144691$

lower p = $\frac{\text{lower } F}{\text{upper } A}$

$= \frac{25.135}{4.295}$

$= 5.852153667$

Both round to 5.9 (2sf / 1dp)

5.9 Nm^{-2}

(Total for question 11 is 3 marks)

12

F = 20.81 N correct to 2 decimal places
P = 5.12 Nm^{-2} correct to 3 significant figures

$$p = \frac{F}{A}$$

p = pressure
F = force
A = area

By considering bounds, work out the value of A to a suitable degree of accuracy.
Give a reason for your answer.

$A = \frac{F}{P}$

F

20.80 ↑ 20.81 ↑ 20.82

20.805 20.815

P

5.11 ↑ 5.12 ↑ 5.13

5.115 5.125

upper A = $\frac{20.815}{5.115}$

$= 4.069403715$

lower A = $\frac{20.805}{5.125}$

$= 4.059512195$

Both round to 4.1 (2sf / 1dp)

4.1 m^2

(Total for question 12 is 3 marks)

13

$$v^2 = u^2 + 2as$$

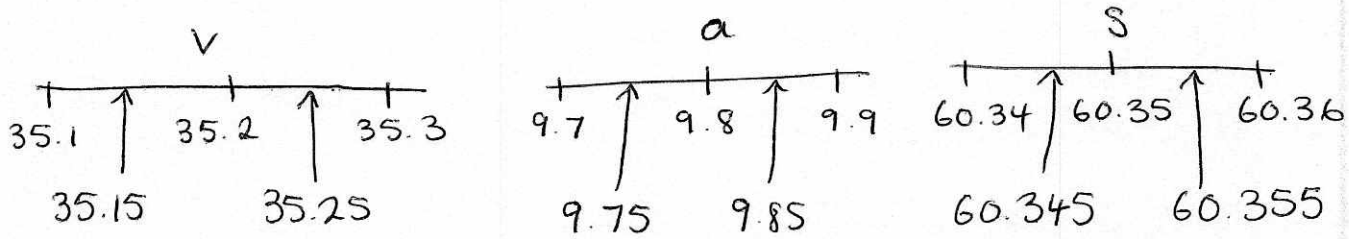
v = 35.2 correct to 1 decimal place
a = 9.8 correct to 1 decimal place
s = 60.35 correct to 2 decimal places

Work out the upper bound for u.
Give your answer to 3 significant figures.

$$v^2 - 2as = u^2$$

$$\sqrt{v^2 - 2as} = u$$

$$\text{upper } u = \sqrt{(\text{upper } v)^2 - 2(\text{lower } a)(\text{lower } s)}$$



$$\text{upper } u = \sqrt{(35.25)^2 - 2(9.75)(60.345)}$$

$$= 8.113877002$$

$$= 8.11 \text{ (3sf)}$$

8.11

(Total for question 13 is 5 marks)